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JavaScript Generators

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A JavaScript Function can only return **one value**.

A JavaScript Generator can **return multiple values**, one by one.

A JavaScript Generator can yield a **stream of data**.

A JavaScript Generator can be **paused** and **resumed**.

Generator Functions

A Generator Function is defined using the **function*** syntax:

```
function* myGenerator() {}
```

When called, a generator function returns a **Generator Object**, not a direct value:

```
{value:1, done:false}
```

Generator Objects

A **Generator Object** is returned by the generator function.

It conforms to both the iterable and iterator protocols, meaning it can be iterated over using **for...of loops**.

```
function* myStream() {
// return {value:1, done:false}
yield 1;

// return {value:2, done:false}
yield 2;

// return {value:3, done:true}
return 3;
}

// Create a Generator
let myGenerator = myStream();

// Iterate over the Generator
for (let value of myGenerator) { // code }
```

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The yield Keyword

The **yield** keyword pauses execution and **yield a value** back to the caller.

The generator's state is preserved, and can be **resumed** from the point of the yield when **next()** is called.

Generators maintain their internal state between yield calls, allowing them to continue execution from where they left off.

Note

The code example above returns 1 and 2. Not 3!

The for..of loop exits when **done:true**.

To get all the values, return them with yield.

Example

```
// return {value:2, done:false}
yield 2;

// return {value:3, done:false}
yeald 3;
}

// Create a Generator
let myGenerator = myStream();

// Iterate over the Generator
for (let value of myGenerator) { // code }
```

Try it Yourself »

Generator Object Methods

A Generator Object conforms to the **iterable and iterator protocols**.

It can be iterated over with **for...of loops**.

It also has **methods**:

Method	Description
next()	Resumes the next execution of a generator function
return()	Finishes this execution and returns a given value
throw()	Throws an exeption in this execution

The next() Method

When called on the Generator object it **resumes** execution of the generator function.

It returns an object with and value and a boolean, indicating if the generator has finished:

{ **value**: *value*, **done**: `true|false` }

They can efficiently generate values on demand, making them suitable for potentially **infinite data streams**.

They can **pause and resume** execution, which can be useful for fine-grained control over program flow.

Before async/await, generators were used with Promises to manage asynchronous operations in a more sequential-looking manner.

Note

Generators bring the **iteration concept** directly into the core JavaScript language and provide a mechanism for customizing the behavior of for...of.

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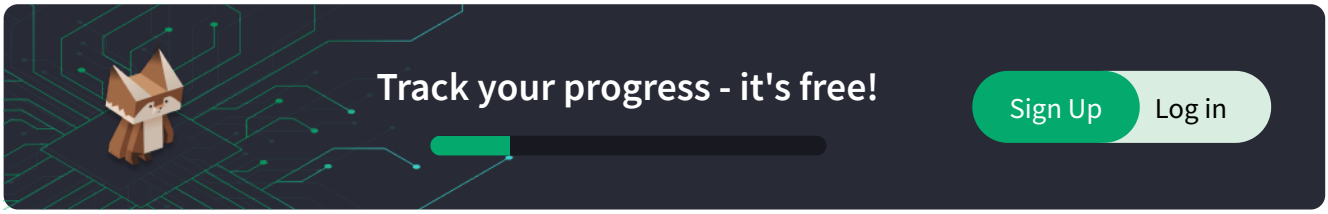

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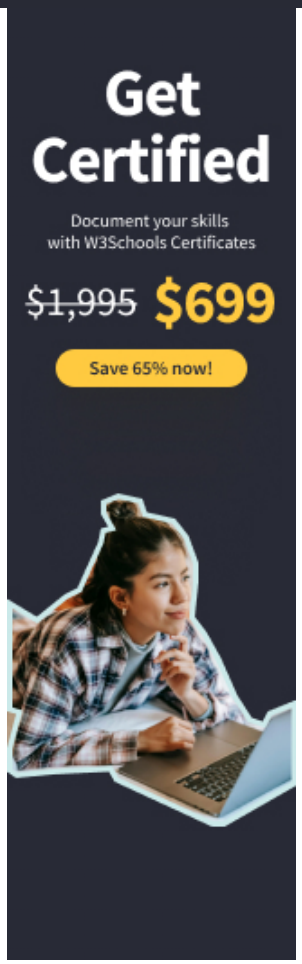

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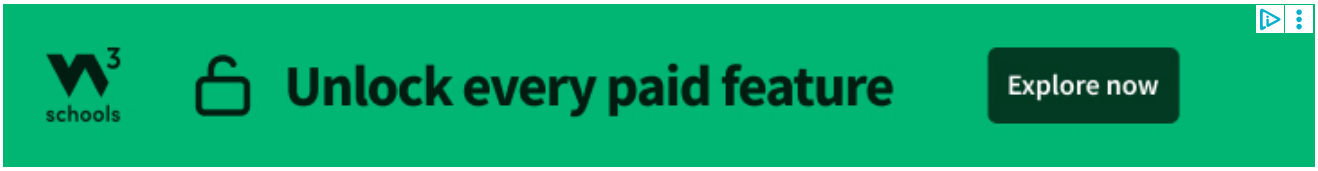

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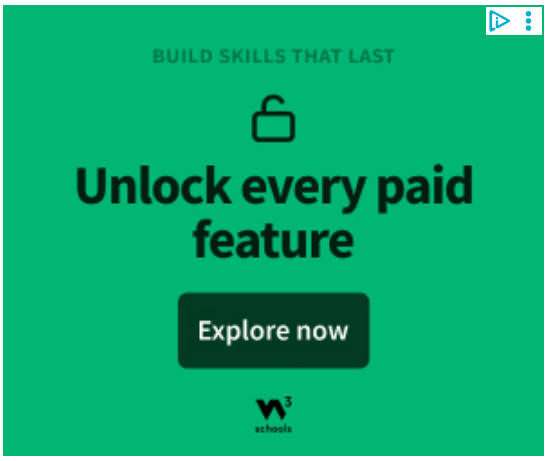

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